

Qatar University
College of Engineering
Mechanical and Industrial Engineering Department

MECH 322 – Mechanical Vibrations
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Progress Report 1

*“Design, Model, Analyze, Build and Test a
Prototype Vibration Absorber”*

Name	QU ID
Mohammed Taha	202309234
Mohammed Elsherbini	202307841
Yousef Hegazi	202307935

Course Instructor: Dr. Mohammad Roshun Paurobally
Group: 8
Beam Specification: Brass, 900 mm × 40 mm × 1 mm
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1 Introduction

A vibration absorber is a passive mechanical system used to reduce unwanted vibrations in structures and machinery. The principle relies on introducing an auxiliary mass–spring–damper system that redistributes the system’s natural frequencies to minimize response at the excitation frequency. Applications range from power transmission lines to tall buildings in earthquake-prone regions.

1.1 Vibration Absorbers Background

Vibration absorbers are secondary mass–spring (and sometimes damping) systems attached to a primary structure to reduce unwanted vibrations by operating on the principle of dynamic vibration absorption, where the absorber oscillates out of phase with the main system to cancel motion at a targeted excitation frequency [1].

When properly tuned, the combined system exhibits two new natural frequencies around the excitation frequency, significantly lowering resonance amplitude. There are several types, including:

- Undamped tuned mass absorbers (effective at a single frequency)
- Damped tuned mass dampers (TMDs) that broaden the operating range
- Adaptive systems that adjust in real time

Historically, vibration absorbers have been widely applied in engineering systems such as:

- Transmission lines to prevent wind-induced oscillations [2]
- Bridges (e.g., Millennium Bridge) to control pedestrian vibrations [1]
- Tall buildings (e.g., Taipei 101) for seismic and wind resistance [3]
- Machinery and vehicle suspensions to reduce noise and improve performance

Their design is typically based on matching the absorber’s natural frequency to the excitation frequency, with damping optimized to balance vibration reduction and system stability.

1.2 Objectives

- Design and prototype a functional vibration absorber for a cantilever beam system.
- Establish baseline characteristics of the unabsorbed system through FEA and mathematical modeling.
- Validate theoretical predictions through experimental testing.
- Demonstrate the effectiveness of vibration absorber design principles in practice.

2 Theory

The two-degree-of-freedom (2-DOF) system with an absorber can be analyzed using frequency response techniques. The undamped absorber is designed such that the absorber

frequency equals the excitation frequency:

$$\omega_a = \sqrt{\frac{k_a}{m_a}} = \omega \quad (1)$$

where:

- ω_a is the absorber natural frequency
- k_a is the absorber stiffness
- m_a is the absorber mass
- ω is the excitation frequency

Damping in the absorber is carefully tuned to broaden the effectiveness while maintaining acceptable amplitudes at the new resonance peaks.

3 System Properties (Group 8)

Table 1. Beam Properties

Property	Value
Length	900 mm
Width	40 mm
Thickness	1 mm
Material	Brass
Young's Modulus	1×10^{11} Pa
Density	8500 kg/m ³
Poisson's Ratio	0.33
Yield Strength	2.39×10^8 Pa

4 Mathematical Model

The cantilever beam system is approximated as a single-degree-of-freedom (SDOF) system consisting of a lumped mass m_1 at the free end connected to the fixed support by an equivalent spring with stiffness k_1 .

4.1 Equation of Motion

$$m_1 \ddot{x}_1 + k_1 x_1 = 0 \quad (2)$$

4.2 Natural Frequency

$$\omega_1 = \sqrt{\frac{k_1}{m_1}} \quad (3)$$

4.3 Equivalent Stiffness

$$k_1 = \frac{3EI}{L^3} \quad (4)$$

where:

- E is Young's modulus
- I is the second moment of area
- L is the beam length

4.4 Beam Properties

Beam mass:

$$m = \rho \cdot V = \rho(LWT) \quad (5)$$

Second moment of area (rectangular section):

$$I = \frac{WT^3}{12} \quad (6)$$

These expressions are used to determine the equivalent stiffness and natural frequency of the system.

5 MATLAB Simulation

5.1 Simulation Setup

The MATLAB model was developed to compute the natural frequency and frequency response of the system using the derived equations.

```

1 clear
2 clc
3 close all
4
5 % Beam properties
6 L = 0.9;           % m
7 E = 1e11;          % Pa
8 b = 0.04;          % m
9 h = 0.001;         % m
10 I = (b*h^3)/12;    % m^4
11

```

```
12 % Masses
13 m_motor = 0.094296;
14 m_cantilever_eff = 0.072216;
15 m_eff = m_cantilever_eff + m_motor;
16
17 % Equivalent stiffness
18 k_eq = (3*E*I)/(L^3);
19
20 % Natural frequency
21 omega_n = sqrt(k_eq/m_eff);
22 f_n = omega_n/(2*pi);
23
24 disp('Natural frequency (Hz):')
25 disp(f_n)
26
27 % Forced vibration parameters
28 F0 = 0.1; % N
29 zeta = 0.0005; % damping ratio
30
31 % Damping coefficient from damping ratio
32 c = 2*zeta*m_eff*omega_n;
33
34 disp('Damping coefficient (N.s/m):')
35 disp(c)
36
37 % Resonance frequency
38 omega_r = omega_n*sqrt(1 - 2*zeta^2);
39 f_r = omega_r/(2*pi);
40
41 disp('Resonance frequency (Hz):')
42 disp(f_r)
43
44 % Frequency response
45 omega = linspace(0.01, 3*omega_n, 3000);
46 r = omega/omega_n;
47
48 % Steady-state amplitude
49 X = (F0/k_eq) ./ sqrt((1-r.^2).^2 + (2*zeta*r).^2);
50
51 % Phase angle
52 phi = atan2(2*zeta*r, (1-r.^2));
```

```
53
54 % Peak from response curve
55 [X_max, idx] = max(X);
56 omega_peak = omega(idx);
57 f_peak = omega_peak/(2*pi);
58
59 disp('Peak frequency from amplitude response (Hz):')
60 disp(f_peak)
61
62 disp('Maximum amplitude (m):')
63 disp(X_max)
64
65 % Plot amplitude response
66 figure
67 plot(omega/(2*pi), X, 'LineWidth', 2)
68 grid on
69 xlabel('Excitation Frequency (Hz)')
70 ylabel('Amplitude (m)')
71 title('Forced Vibration Amplitude Response')
72
73 % Time response at resonance
74 t = linspace(0, 40, 4000);
75 x_res = X_max*sin(omega_peak*t - phi(idx));
76
77 figure
78 plot(t, x_res, 'LineWidth', 1.5)
79 grid on
80 xlabel('Time (s)')
81 ylabel('Displacement (m)')
82 title('Steady-State Response at Resonance')
```

Listing 1. MATLAB script for 2-DOF vibration absorber frequency response – Group 13.

6 MATLAB Results

6.1 Natural Frequency

$$f_n = 0.4568 \text{ Hz}$$

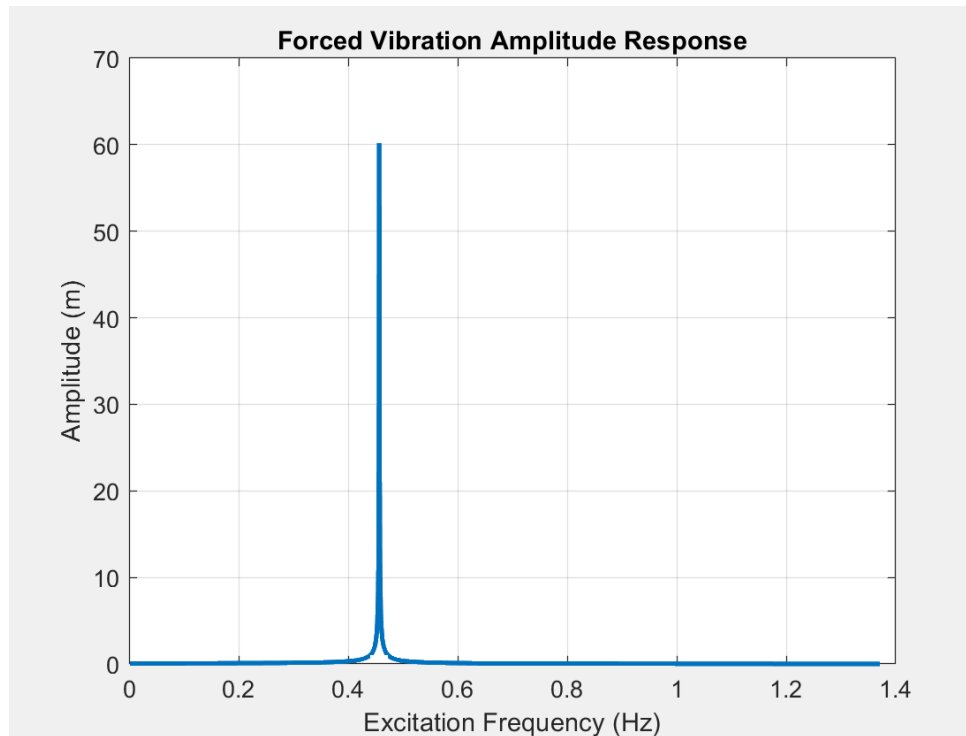


Figure 1. Frequency Response Curve

7 SolidWorks Simulation

Finite Element Analysis (FEA) was conducted using SolidWorks to obtain more accurate natural frequencies and mode shapes.

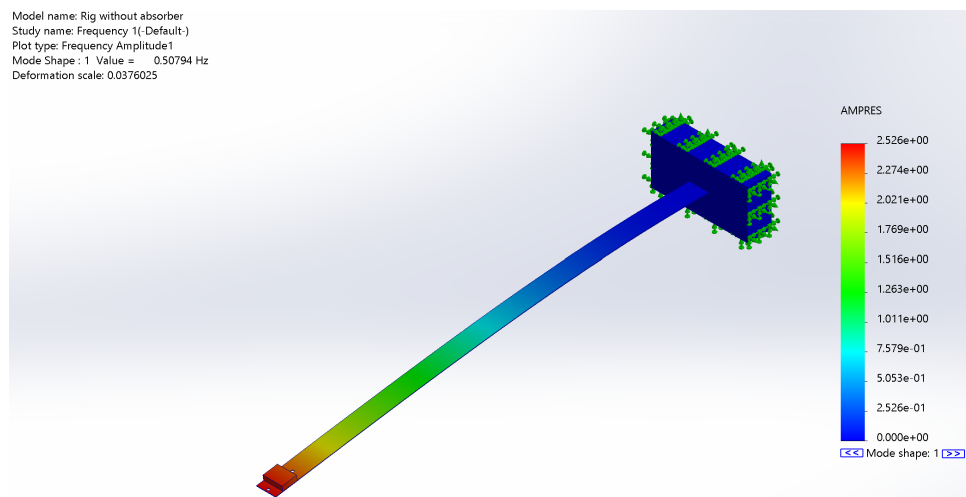


Figure 2. With Motor Mass - Free Vibration Natural Frequency Mode 1

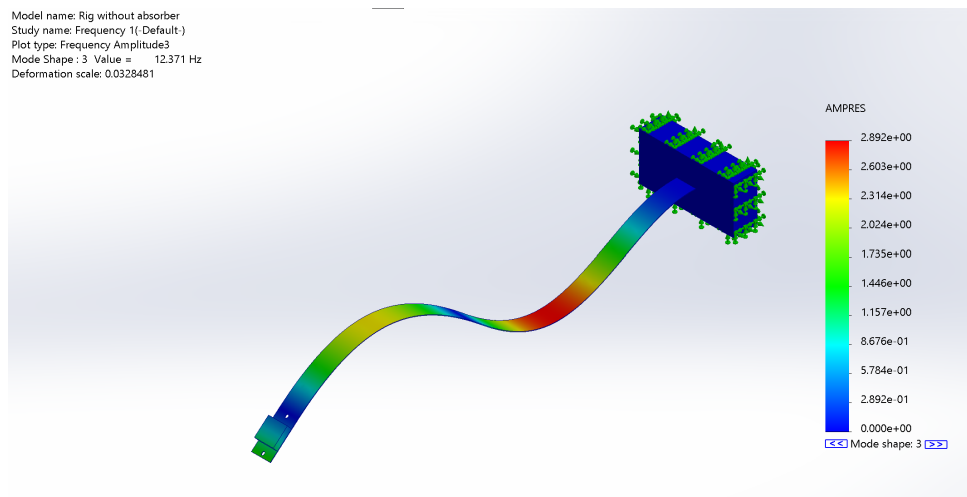


Figure 4. With Motor Mass - Free Vibration Natural Frequency Mode 3

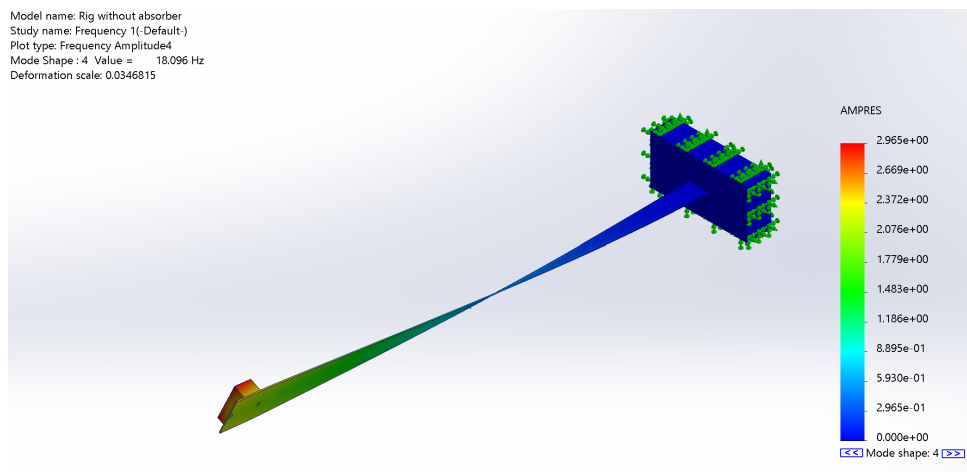


Figure 5. With Motor Mass - Free Vibration Natural Frequency Mode 4

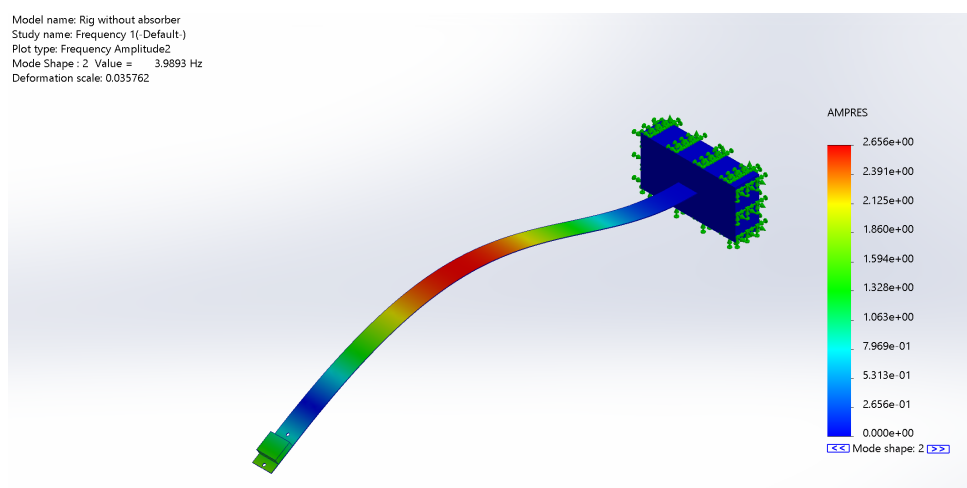


Figure 3. With Motor Mass - Free Vibration Natural Frequency Mode 2

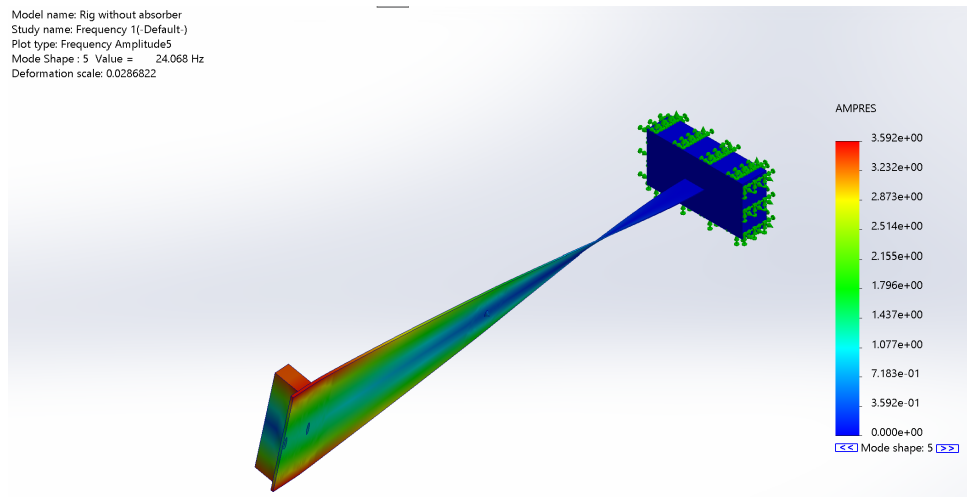


Figure 6. With Motor Mass - Free Vibration Natural Frequency Mode 5

7.1 Natural Frequencies

Table 2. Mode Frequencies

Mode	Frequency (Hz)
1	0.50794
2	3.9893
3	12.371
4	18.096
5	24.068

8 Comparison

Comparison between MATLAB and SolidWorks natural frequencies:

- MATLAB: 0.4568 Hz
- SolidWorks (Mode 1): 0.50794 Hz

The natural frequency obtained from MATLAB (0.4568 Hz) is slightly lower than the first-mode natural frequency from SolidWorks (0.50794 Hz), showing relatively close agreement.

This difference can be attributed to:

- Simplifications in the MATLAB model (ideal boundary conditions, lumped parameters)
- Neglect of geometric details
- Finite element modeling effects in SolidWorks
- Differences in mass and stiffness estimation

- Meshing and boundary condition implementation

Overall, the results are reasonably consistent, indicating that the MATLAB model provides a good approximation of the system's dynamic behavior.

9 Appendix – Minutes of Meetings

Minutes of Meeting 1: Project Planning & System Definition

Date: 10 February 2026

Agenda:

1. Understand project requirements for vibration absorber system
2. Assign roles and responsibilities among team members
3. Select beam specifications for Group 8
4. Plan initial modelling approach (SolidWorks & MATLAB)

Discussion:

The team reviewed the project brief and discussed the main objective, which is to design and analyze a vibration absorber system for a cantilever beam.

It was confirmed that Group 8 will use a **brass cantilever beam with dimensions 900 mm × 40 mm × 1 mm**.

The team agreed to begin with:

- SolidWorks modelling to obtain natural frequencies and mode shapes
- MATLAB modelling assuming a single degree of freedom (SDOF) system

Responsibilities were divided as follows:

- Mohammed Elsherbini: SolidWorks modelling and modal analysis
- Yousef: MATLAB modelling and analytical calculations
- Mohammed Taha: Report writing, literature review, and documentation

Action Items:

1. Create SolidWorks model of the cantilever beam
2. Extract first 5 natural frequencies and mode shapes
3. Start MATLAB SDOF model for comparison
4. Begin literature review on vibration absorbers

Minutes of Meeting 2: Simulation Results & Model Validation

Date: 18 February 2026

Agenda:

1. Review SolidWorks modal analysis results
2. Compare MATLAB results with simulation
3. Discuss assumptions used in modelling
4. Prepare content for Progress Report 1

Discussion:

The team reviewed the SolidWorks modal analysis results. The first natural frequency of the beam was identified and the first five mode shapes were obtained.

MATLAB modelling was completed assuming an equivalent mass-spring system. The calculated natural frequency showed good agreement with the SolidWorks results, confirming the validity of the analytical model.

The team discussed the main assumptions used:

- Beam modeled as SDOF system
- Small damping neglected
- Motor mass represented as lumped mass

It was agreed that both models are consistent and can be used for further absorber design in Progress Report 2.

Action Items:

1. Finalize MATLAB plots (amplitude and phase response)
2. Insert SolidWorks mode shape figures in report
3. Complete literature review section
4. Compile and submit Progress Report 1

References

- [1] Den Hartog, J. P., *Mechanical Vibrations*, 4th ed., McGraw-Hill, 1956.
- [2] Irvine, H. M., *Cable Structures*, MIT Press, 1981.
- [3] Soong, T. T., and Dargush, G. F., *Passive Energy Dissipation Systems in Structural Engineering*, Wiley, 1997.